

Davide Giovagnoli

E-mail : d.giovagnoli [at] unibo.it
website - Scholar

Education

PhD in Mathematics — Bologna, Italy
Alma Mater Studiorum - Università di Bologna — Nov 2023 - Nov 2026

- Advisor: Prof. Fausto Ferrari
- Research interests: Regularity theory of Nonlocal/NonlinearElliptic/Parabolic PDEs, Free Boundary Problems, Calculus of Variations, and Analysis on Carnot groups.

Master's Degree in Mathematics — Bologna, Italy
Alma Mater Studiorum - Università di Bologna — Oct 2021 - Sep 2023

- Thesis title: Alt-Caffarelli-Friedman monotonicity formulas, Supervisor: Fausto Ferrari
- Final Grade: 110/110 cum laude (weighted grade average: 29.93/30)

Bachelor's Degree in Mathematics — Bologna, Italy
Alma Mater Studiorum - Università di Bologna — Sep 2018 - Oct 2021

- Thesis title: The Fractional Laplacian defined through an analytical extension, Supervisor: Fausto Ferrari
- Final Grade: 110/110 cum laude (weighted grade average: 29.07/30)

Articles and preprints

1. **D. Giovagnoli** & **D. Jesus** , *Hölder Gradient Estimates for (s,p)-Poisson Equations with measurable and Hölder Data,* preprint arXiv:2610.12012
2. **G. Csató** & **D. Giovagnoli**, *Continuity and monotonicity of weighted perimeters of convex sets*, preprint arXiv:2610.11677
3. **D. Giovagnoli** & **D. Jesus** & **L. Silvestre**, *Hölder gradient estimates for fractional p-caloric functions,* preprint arXiv:2609.20097
4. **F. Ferrari** & **D. Giovagnoli** & **D. Jesus**, *A transmission problem arising from the two-phase Stefan problem*, preprint arXiv:2608.24734
5. **G. Csató** & **D. Giovagnoli** & **P. Roy,** *Weighted Perimeters and pth Moments of Inertia of Convex Curves and Surfaces*, preprint arXiv:2608.19851
6. **D. Giovagnoli** & **D. Jesus** & **L. Silvestre** *$C^{1,\alpha}$ regularity for fractional p-harmonic functions,* preprint arXiv:2509.26565
7. **F. Ferrari** & **D. Giovagnoli** & **E. M. Merlino**, *On a fractional Alt-Caffarelli-Friedman-type monotonicity formula,* preprint, arXiv:2509.25891
8. **F. Ferrari** & **D. Giovagnoli** & **D. Jesus**, *On the Geometry of Solutions of the Fully Nonlinear inhomogeneous one-phase Stefan problem,* preprint arXiv:2504.12912
9. **F. Ferrari** & **N. Forcillo** & **D. Giovagnoli** & **D. Jesus**, *Free boundary regularity for the inhomogeneous one-phase Stefan problem,* Commun. Contemp. Math. (2026).
10. **D. Giovagnoli** & **E. M. Merlino** & **D. Moreira** *A quantitative Hopf-Oleinik lemma for degenerate fully nonlinear operators and applications to free boundary problems,* La Matematica, 5, 22 (2026).
11. **D. Giovagnoli** & **D. Jesus**, *A Fully Nonlinear Transmission problem degenerating on the interface,* Calc. Var. Partial Differential Equations 65, 93 (2026).
12. **F. Ferrari** & **D. Giovagnoli**, *Some counterexamples to Alt-Caffarelli-Friedman monotonicity formulas in Carnot groups,* Ann. Mat. Pura Appl. 204, 427–445 (2025).

Books and surveys

1. **D. Giovagnoli** & **D. Jesus**, *Regularity results in elliptic and parabolic transmission problems*, survey, Bruno Pini Math. Anal. Semin., Vol. 16, No. 1, 168-188 (2025).
2. **D. Giovagnoli** & **M. Talluri** & **J.M. Urbano** *A note on the rigidity of solutions to parabolic fully nonlinear degenerate equations*,
3. **D. Giovagnoli** & **E. Salato**, *A gentle introduction to the Alt-Caffarelli-Friedman monotonicity formula,* Book draft in preparation

Talks & Presentations

S.P.I.E.Dini Seminar, University of Florence — Florence, Italy
Seminar talk — September 24, 2026

Workshop on Nonlinear PDEs, Harmonic Analysis and FBPs at KTH, — Stockholm, Sweden
Invited talk — August 17-19, 2026

The 15th AIMS Conference 2026 — Athens, Greece
Invited talk in Special session: Regularity in local versus nonlocal problems
Contributed talk for AIMS Student Paper Competition 2026 — July 6-10, 2026

3rd Meeting UMI for Doctoral Students — Florence, Italy
Contributed talk — June 29, 2026

(Ir)Regularity @Parma I — Parma, Italy
Poster presentation — May 18-22, 2026

Bruno Pini Analysis Seminar, University of Bologna — Bologna, Italy
Seminar talk — May 7, 2026

Analysis Seminar, University of Milan — Milano, Italy
Seminar talk — March 31, 2026

Calderón-Zygmund Analysis Seminar, University of Chicago — Chicago, USA
Seminar talk — February 27, 2026

Seminari d'Anàlisi de Barcelona, Universitat de Barcelona — Barcelona, Spain
Seminar talk — July 10, 2025

21st JISD 2025 Summer School, CRM — Barcelona, Spain
Poster presentation — June 30, 2025

PDE and Analysis Seminar, University of Pittsburgh — Pittsburgh, USA
Seminar talk — March 24, 2025

Free boundaries in action (in honor of Sandro Salsa) — Bologna, Italy
Invited talk — March 11, 2025

PDE Geometric Analysis Seminar, University of Wisconsin-Madison — Madison, USA
Seminar talk — March 10, 2025

Luis Silvestre's Research Group Seminar, University of Chicago — Chicago, USA
Seminar talk — February 25, 2025

ASK Seminar, University of Bologna — Bologna, Italy
Seminar talk — January 14, 2025

Latinos on free boundaries in Bologna — Bologna, Italy
Short communication — July 29, 2024

SHaCAMiGa — Bologna, Italy
Contributed talk — January 25, 2024

Research visits

University of Chicago | Chicago, USA — February - March 2026 (2 weeks)
Invited by Prof. L. Silvestre

University of Wisconsin | Madison, USA — February - March 2026 (3 weeks)
Invited by Prof. M. Feldman

Universitat de Barcelona | Barcelona, Spain — January 2026 (3 weeks)
Invited by Prof. G. Csato

Universitat de Barcelona | Barcelona, Spain — July 2025 (2 weeks)
Invited by Prof. G. Csato

Columbia University | New York, USA — April 2025 (1 week)
Invited by Prof. D. De Silva and Prof. O. Savin

University of Pittsburgh | Pittsburgh, USA — March 2025 (1 week)
Invited by Prof. J. Manfredi

University of Wisconsin | Madison, USA — March 2025 (3 weeks)
Invited by Prof. M. Feldman

University of Chicago | Chicago, USA — February 2025 (2 weeks)
Invited by Prof. L. Silvestre

Universitat de Barcelona | Barcelona, Spain — May - June 2024 (1 month)
Invited by Prof. G. Csato

Universitat de Barcelona | Barcelona, Spain — July 2023 (3 weeks)
Invited by Prof. G. Csato

Teaching

- **Teaching assistant for "Analisi Matematica 2",** Sep 2025 - Feb 2026
  Calculus II for second year students in Mathematics taught by F. Ferrari, University of Bologna
- **TA for "Analisi Matematica 2",** Sep 2025 - Feb 2026
  Calculus II for second year students in Astronomy taught by M.C. Tesi, University of Bologna
- **TA for "Analisi Matematica T-B",** Feb - Sep 2025
  Calculus II for first year students in Energy Engineering taught by F. Ferrari and E. Vecchi, University of Bologna
- **TA for "Analisi Matematica T-B",** Feb - Sep 2024
  Calculus II for first year students in Energy Engineering taught by F. Ferrari, University of Bologna
- **TA for "Advanced Mathematical Analysis",** Oct 2023 - Feb 2024
  Mathematics Master's course taught by N. Arcozzi, University of Bologna
- **TA for the project "AlmaMathematica",** Feb 2022 - Oct 2023
  Series of online seminars aimed at high school students for the TOLC exam (Test Online for the skills assessment) under supervision of G. Cupini, University of Bologna
- **Mathematics teacher for summer remedial courses** June - July 2021
  Review lessons and practice exercises for second, third, and fourth grades to recover school debt, Liceo Scientifico Malpighi Bologna

Editorial Service

Peer reviewer for: Proc. Amer. Math. Soc., Nonlinear Anal., Adv. Calc. Var., Comm. in PDEs., J. Geom. Anal.

Projects and funds

I am involved in the following projects:

- Member of the PRIN 2022 Project 20227HX33Z Nov 2023 - Sep 2025
  Pattern formation in nonlinear phenomena **CUP J53D23003610006**
  Principal Investigator: Susanna Terracini.
- Member of the INdAM - GNAMPA Project 2025, CUP E5324001950001:
  At the edge of Ellipticity. Principal Investigator: I. Birindelli.
- INCENTIVI PER COLLABORAZIONI CON UNIVERSITÀ DEL NORD AMERICA 2024 MAT: Interplaying problems in analysis and geometry.
  Coordinated by F. Ferrari.
- INCENTIVI PER COLLABORAZIONI CON UNIVERSITÀ DEL NORD AMERICA 2025 MAT: Free Boundaries Problems: Analytical and Numerical Aspects.
  Coordinated by F. Ferrari.

Skills

Languages: Italian (Native speaker), English (Advanced level), Spanish (Basic level).

English C1 level CLA Certificate *June 2025*
CLA Academics LEA Course C2 level English Course (50 hrs) *Feb 2025 - June 2025*
CLA Academics LEA Course C1 level English Course (50 hrs) *Oct 2024 - Jan 2025*

Programming: LaTeX, MATLAB, Python.

Other scientific activities

Finalist for Student Paper Competition 2026 at AIMS 2026

Organizer of ASK seminars (Seminar website)
A cycle of seminars and activities for young Analysis researchers in Bologna (>30 seminars held)

Organizer of ASK Conference 2026 (Conference website)
Conference in Geometric Analysis and PDEs, University of Bologna September 2 - 4, 2026

Summer School JISD 2025
21th School on Interactions between Dynamical Systems and Partial Differential Equations at CRM, Barcelona, June 30 - July 4, 2025

Organizer of ASK Conference 2024 (Conference website)
Interaction between different areas of Mathematical Analysis and Probability, University of Bologna December 18 - 20, 2024

SIIT-LSU Conference on Analysis and PDE Online, September 26, 2024

AMS - UMI International Joint Meeting Palermo, July 23 - 26, 2024

First JHU - UNIBO Conference "Noncommutativity at the Interface of Topology, Geometry and Analysis" Bologna, June 24 - 28, 2024

Satellite ECM 2024 Conference "BAC24" Barcelona, June 3 - 7, 2024

Workshop "Highlights in Nonlinear PDEs" Gaeta, April 8-11, 2024

Workshop "CalVarFer" Ferrara, March 6-7, 2024

Workshop on "Regularity and geometric aspects of nonlinear PDEs" Pisa, January 31 - February 2, 2024

Summer School JISD 2023
19th School on Interactions between Dynamical Systems and Partial Differential Equations at CRM, Barcelona, July 3 - 7, 2023

Conference "NonPUB 2023 - Nonlocal and Nonlinear Partial Differential Equations" Bologna, June 8-9, 2023

PhD Courses

"Free Boundary Problems for Elliptic Equations and Shock Analysis"
taught by M. Feldman (University of Wisconsin-Madison), June 2026

"Degenerate or Singular Elliptic and Parabolic Equations"
taught by P. Ambrosio (Unibo), March - April 2026

"Second Order Parabolic Equations" taught by S. Ciani (Unibo), Dec 2025

"Harmonic mappings with free boundaries and their heat flows"
taught by Y. Sire (JHU), Nov - Dec 2025

"Fundamental Properties of Harmonic Functions" taught by D. Moreira (UFC), Sep 2025

"Qualitative properties of solutions of p-Laplace equations"
taught by E. Vecchi (Unibo), and S. Biagi (Polimi) May 2025

"Fully nonlinear elliptic equations" taught by D. Jesus (Unibo), Jan - Feb 2025

"Introduction to the regularity up to the boundary for the Bernoulli type free boundary problems" taught by D. Moreira (UFC), Sep - Oct 2024

"An invitation to nonlinear Regularity Theory: the stationary and diffusive p-Laplacean"
taught by S. Ciani (Unibo), June - July 2024

"The Heat Equation: Dirichlet and Cauchy problems"
taught by E. Lanconelli (Unibo), Apr - Sep 2024

"First order nonlinear PDEs: from theory to approximation"
taught by P. Albano (Unibo) and S. Tozza (Unibo), May 2024

"Large solutions for semilinear PDEs" taught by N. Abatangelo (Unibo), Mar - Apr 2024

"Fine properties of functions of bounded variation and sets of finite perimeter"
taught by G.E. Comi (Unibo), Mar - Apr 2024

"Diffraction Theory and Complex Analysis in Several Variables"
taught by V. Kunz (OSU), Dec 2023 - Jan 2024

"Viscosity solutions of second order elliptic equations and applications"
taught by S. Salsa (Polimi) and F. Ferrari (Unibo), Feb - March 2023

I authorize the processing of personal data contained in my curriculum vitae.

Bologna, October 12, 2026 Davide Giovagnoli